Unit PoE CAM-W

SKU:U121-B















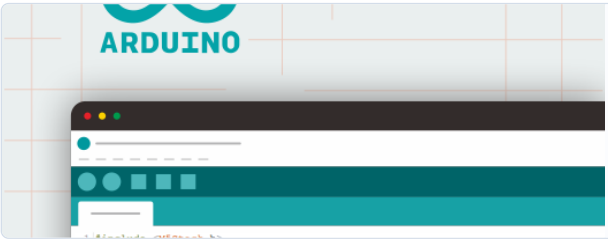




Description

Unit PoE CAM-W is a programmable PoE camera based on the "ESP32 + W5500" solution. It integrates PoE Ethernet communication and power-over-Ethernet functionality, along with Wi-Fi capabilities, and includes a 2MP image sensor OV2640. This product is suitable for image monitoring, remote data collection, and scenarios requiring PoE wiring.

Tutorial



Arduino IDE

This tutorial will introduce you to how to program and control the Unit PoE CAM-W device using Arduino IDE

Features

- Built-in ESP32 MCU (supports WiFi)
- 2MP camera OV2640
- Ethernet chip W5500
- LAN speed up to 10M or 100M (depending on the provided line)
- PoE IEEE802.3 AF standard (maximum power 6W)
- Built-in indicator LED
- Development platforms: Arduino, ESP-IDF
- Standard Ethernet interface RJ45
- Standard LEGO interface and camera mount

Includes

- 1 x Unit PoE CAM-W
- 1 x Camera mounting clip
- 1 x LEGO-compatible clip

Applications

- Warehouse monitoring
- Timed photography
- Machine vision
- Home smart monitoring

Specifications

Specification	Parameter
MCU	ESP32-D0WDQ6-V3
Camera Module	OV2640
Ethernet Chip	W5500
Flash	16MB
PSRAM	8MB
Pixel Size	2.2 µm x 2.2 µm
Output Format (8-bit)	YUV (422/420) /YCbCr422, RGB565/555, 8-bit compressed data, 8-/10-bit Raw RGB data
Automatic Image Control	Auto Exposure Control (AEC), Auto Gain Control (AGC), Auto White Balance (AWB), Auto Band Filter (ABF), and Auto Black Level Calibration (ABLC)
DFOV	65°
Supported TCP/IP Protocols	TCP, UDP, ICMP, IPv4, ARP, IGMP, PPPoE
Ethernet Interface	RJ45
Maximum Power	6W
Product Size	64 × 24 × 18mm
Package Size	80 × 37 × 32mm
Product Weight	38g
Gross Weight	45g

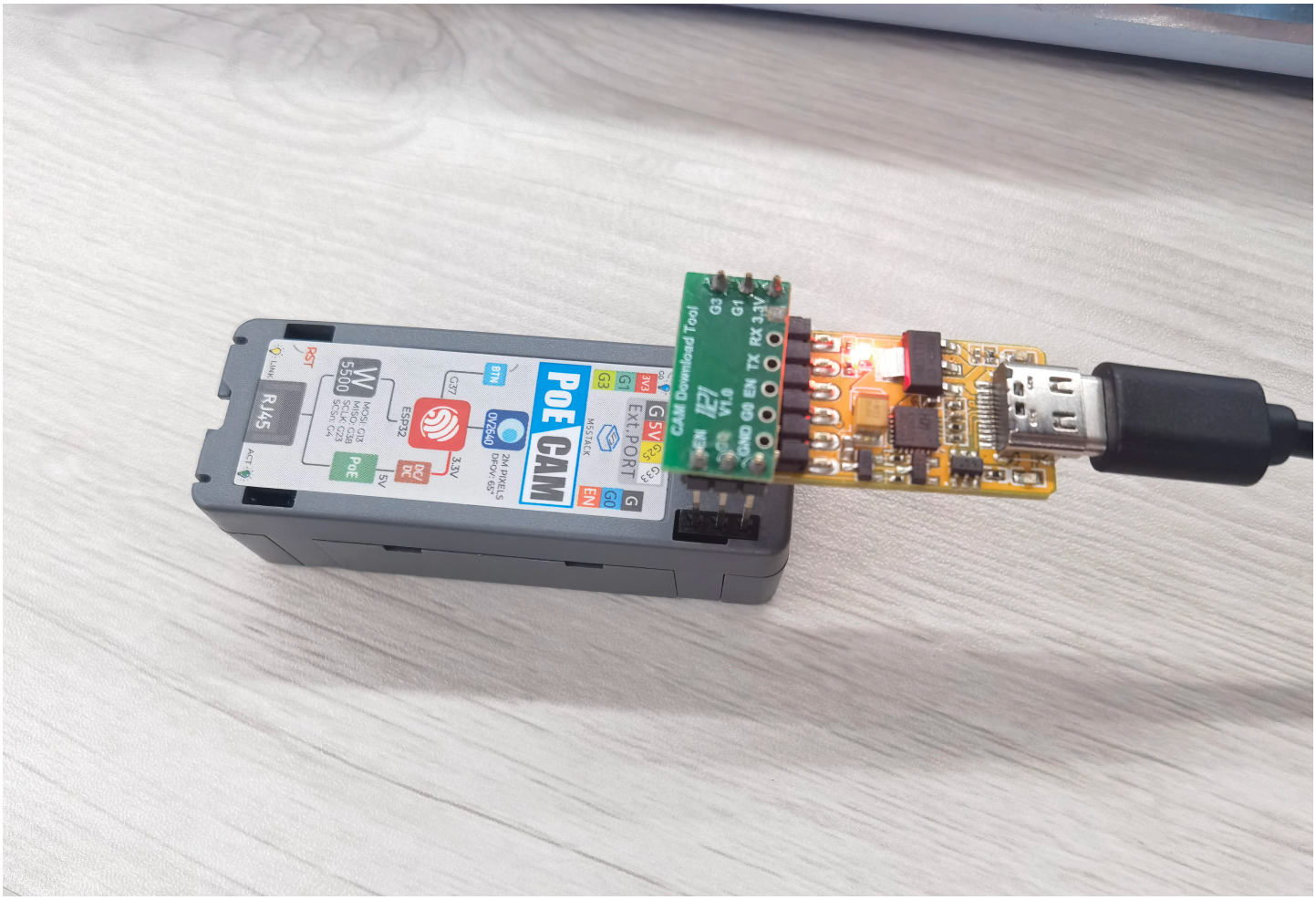
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Notes for Use

Continuous acquisition of high-resolution image streams by the camera may cause the device to overheat and result in abnormal image colors. For prolonged operation, additional cooling is recommended.

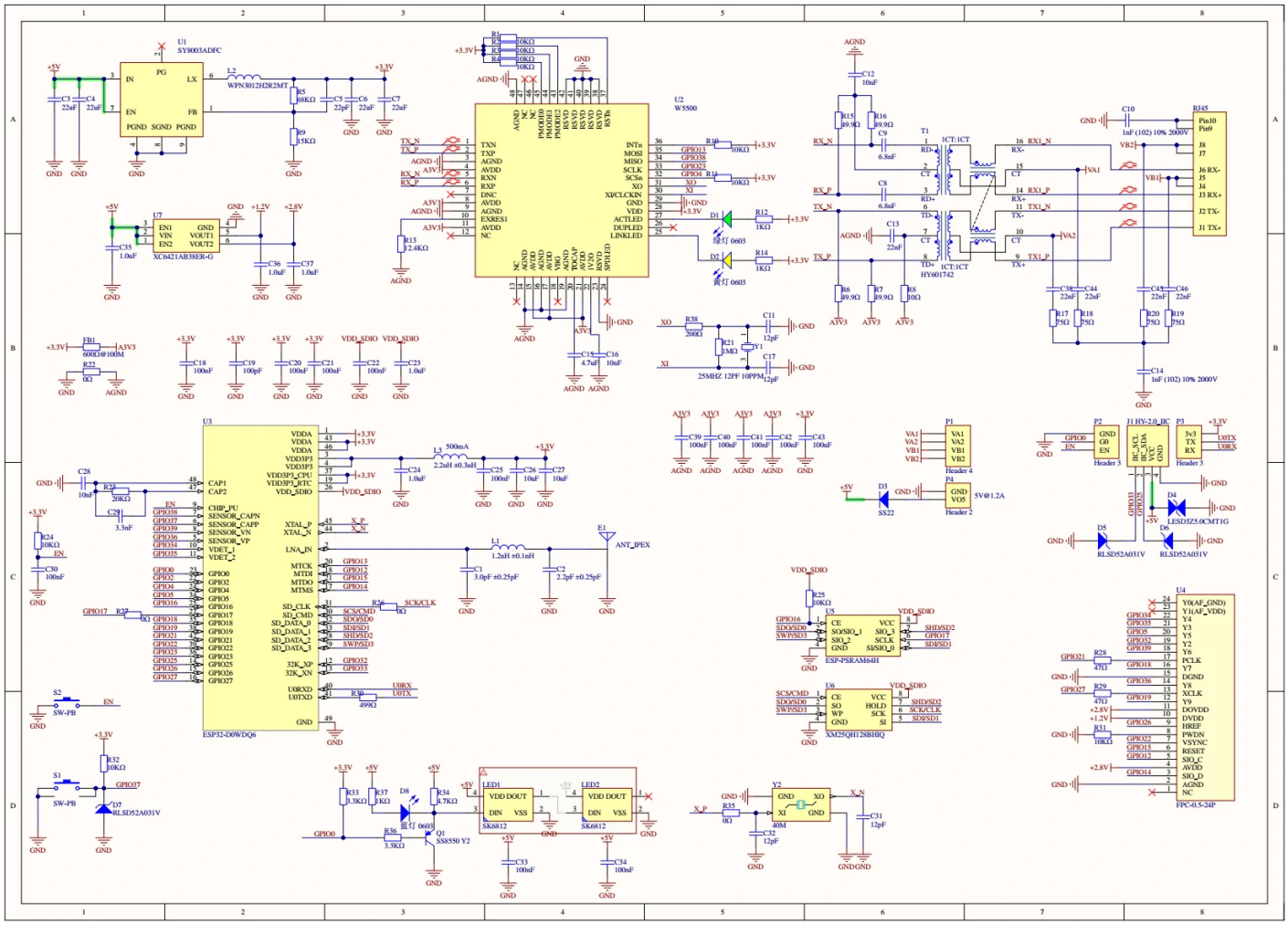
Note:

`When downloading the program for PoECAM, you need to connect an external ESP32 programmer`. You can click here to purchase the official M5 ESP32 Downloader Kit, which includes a convenient connection board



Schematics

- Unit PoE CAM-W Schematic PDF



PinMap

OV2640

Interface	Camera Pin	PoE CAM
SCCB Clock	SIOC	G12
SCCB Data	SIOD	G14
System Clock	XCLK	G27
Vertical Sync	VSYNC	G22
Horizontal Reference	HREF	G26
Pixel Clock	PCLK	G21
Pixel Data Bit 0	D0	G32
Pixel Data Bit 1	D1	G35
Pixel Data Bit 2	D2	G34
Pixel Data Bit 3	D3	G5
Pixel Data Bit 4	D4	G39
Pixel Data Bit 5	D5	G18
Pixel Data Bit 6	D6	G36
Pixel Data Bit 7	D7	G19
Camera Reset	RESET	G15
Camera Power Down	PWDN	-1
Power Supply 3.3V	3V3	3V3
Ground	GND	GND

W5500

W5500	PoE CAM
MOSI	G13
MISO	G38
SCLK	G23
CS	G4

EXT.PORT

EXT.PORT	PoE CAM
RX	G25
TX	G33
5V	5V
GND	GND

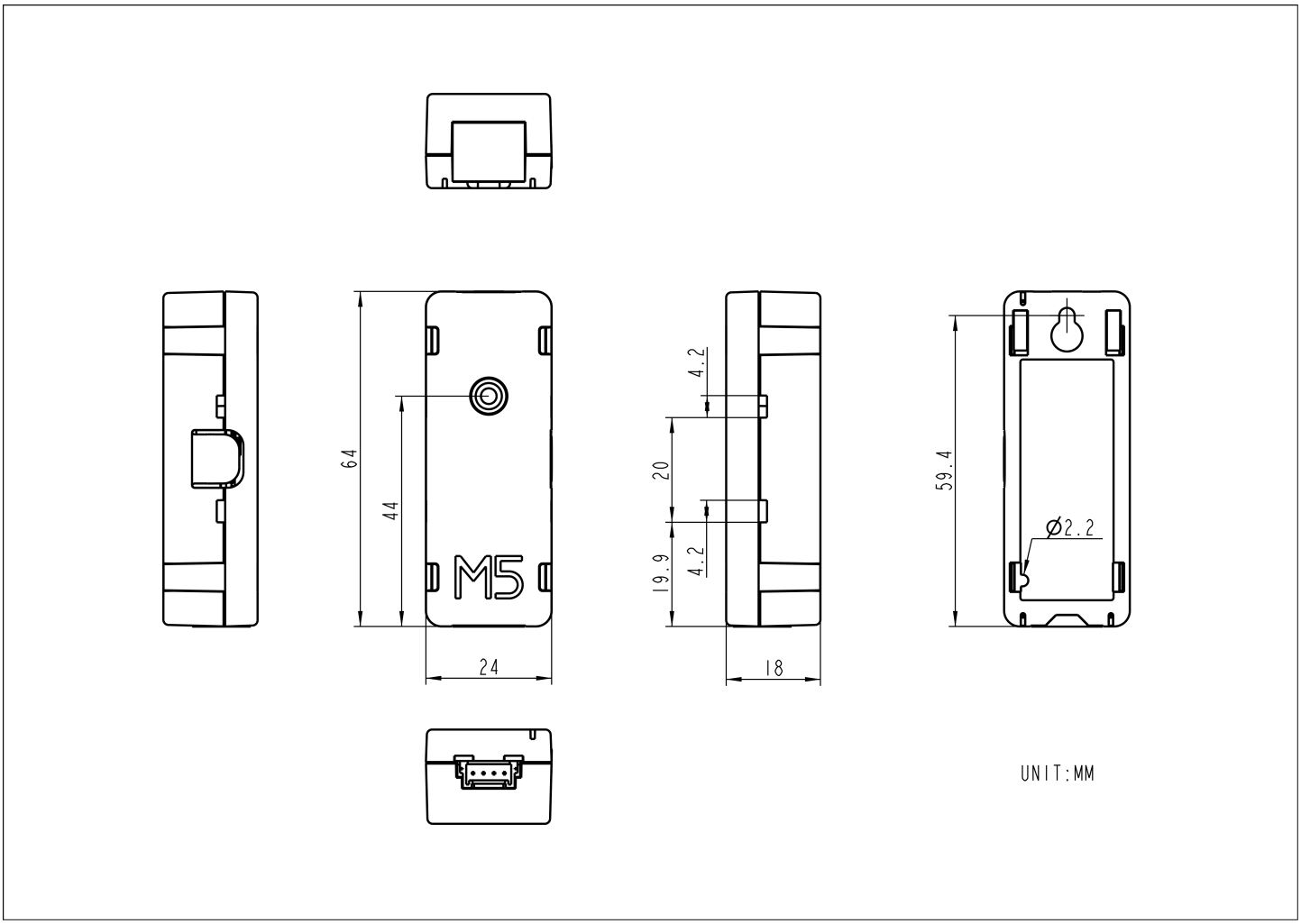
Right Button

Button	PoE CAM
Signal	G37

Blue LED

Blue LED	PoE CAM
Signal	G0

Model Size



Datasheets

- ESP32
- OV2640
- W5500
- Easyloader
- PoECAM-W PinMap

Softwares

Arduino

- Unit PoE CAM-W Arduino Quick Start
- Unit PoE CAM-W Arduino Library
- Unit PoE CAM-W Button Example
- Unit PoE CAM-W Capture Example

- Unit PoE CAM-W HTTP POST Example
- Unit PoE CAM-W LED Control Example
- Unit PoE CAM-W RTSP Example
- Unit PoE CAM-W Web CAM Example

EasyLoader

Easyloader	Download Link	Note
Unit PoE CAM-W Test Easyloader	download	/

Video

PoECAM_VIDEO.mp4